Introduction to Theoretical Computer Science

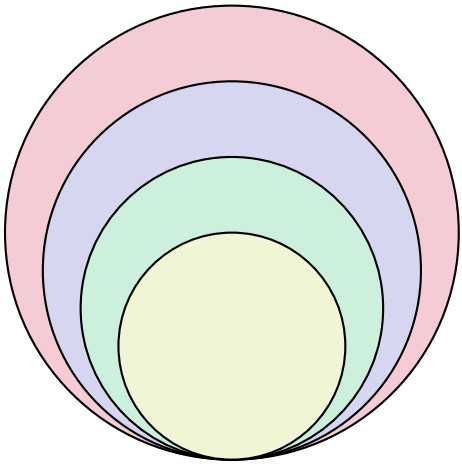
CS2490 - Session 24: Context-Free Grammars

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Chomsky Hierarchy (aka the *onion*)



Everything (including non-computable)

Recursively enumerable languages

Context-free languages

Regular languages

Context-free grammars

A grammar $G = (\Sigma, V, S, P)$ is said to be context-free if all its production rules are of the form:

$$A \to x$$

where $A \in V$ is a single variable, and $x \in (\Sigma \cup V)^*$.

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The name context-free comes from the fact that the variable A can be replaced by x regardless of the context in which it appears.

A language is said to be context-free if it can be generated by a context-free grammar. We use the acronyms CFG for context-free grammar and CFL for context-free language.

Example 1

The grammar $G = (\{a, b\}, \{S\}, S, P)$ with production rules:

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$$\mathrm{L}(G) = \{w \in \{a, b\}^* \mid w = w^R \wedge |w| \text{ even}\}$$

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The grammar $G = (\{0, 1\}, \{S\}, S, P)$ with production rules:

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$$\mathrm{L}(G) = \{w \in \{0,1\}^* \mid n_0(w) = n_1(w)\}$$

Derivations and ambiguity

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Note: The last two have the same *flavor*, compared to the first one.

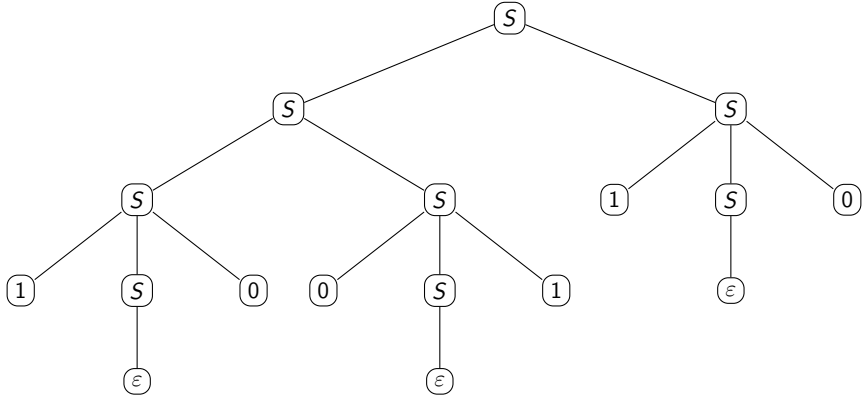
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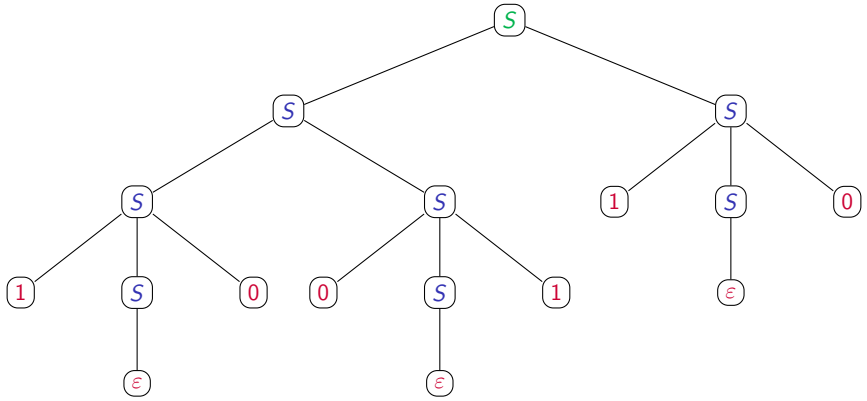
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4. If an internal node is labeled with a variable A and has children labeled with symbols $x_1, x_2, \ldots, x_k$, then there is a production rule in P of the form $A \to x_1 x_2 \ldots x_k$; a child labeled ε must be the only child ($k = 1$).

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Replace condition 1 by "The root is labeled with $A \in V$" and condition 2 by "Each leaf is labeled with a letter, the empty word, *or a variable*", and you get a partial derivation tree rooted in A.

Derivation trees and derivations

Claim:

Let $G = (\Sigma, V, S, P)$ be a context-free grammar. $V \ni A \Rightarrow^* u \in (\Sigma \cup V)^*$ in G if and only if there exists a partial derivation tree rooted in A for G with yield u.

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Consequence: $S \Rightarrow^* w \in \Sigma^*$ if and only if there exists a derivation tree for G with yield w.

Derivation trees and derivations (cont'd)

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($\Leftarrow$) By induction on the height of the tree.

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($\Leftarrow$) By induction on the height of the tree. If the height is 0, then the tree is a single node labeled A and $u = A$.

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($\Leftarrow$) By induction on the height of the tree. If the height is 0, then the tree is a single node labeled A and $u = A$. Otherwise, if the height is $h + 1$, the root is labeled A and has children labeled $x_1, x_2, \ldots, x_k$ such that there is a production rule $A \to x_1 x_2 \ldots x_k$.

Derivation trees and derivations (cont'd)

Proof.
($\Rightarrow$) By induction on the length of the derivation. If the length is 0, then $u = A$ and the tree is a single node labeled A. Otherwise, if the length is $n+1$, the derivation is of the form $A \Rightarrow^* xBy \Rightarrow xvy = u$ where $B \to v$ is a production rule. By induction hypothesis, there exists a partial derivation tree with yield xBy. Replace the leaf corresponding to this occurrence of B by a subtree whose root is labeled B and whose children are labeled by the symbols of v (a single child ε if $v = \varepsilon$). The resulting tree has yield xvy, i.e., u.
($\Leftarrow$) By induction on the height of the tree. If the height is 0, then the tree is a single node labeled A and $u = A$. Otherwise, if the height is $h+1$, the root is labeled A and has children labeled $x_1, x_2, \ldots, x_k$ such that there is a production rule $A \to x_1 x_2 \ldots x_k$. Each child is the root of a subtree of height at most h, so by induction hypothesis (or trivially, if x_i is a letter or ε), there exists a derivation $x_i \Rightarrow^* y_i \in (\Sigma \cup V)^*$ from each child's label x_i to its yield y_i. Concatenating these derivations with the production rule $A \to x_1 x_2 \ldots x_k$ gives a derivation from A to $y_1 y_2 \ldots y_k = u$. $\square$

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$S \Rightarrow SS \Rightarrow SSS \Rightarrow 1S0SS \Rightarrow 1S00S1S \Rightarrow 1S00S11S0 \Rightarrow 1S00S110 \Rightarrow 100S110 \Rightarrow 100110$ is neither a leftmost nor a rightmost derivation.

Derivation trees and leftmost/rightmost derivations

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(The same holds for rightmost derivations.)

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A context-free language is inherently ambiguous if every context-free grammar that generates it is ambiguous.

Example of an ambiguous grammar

Consider the grammar $G = (\{0, 1, \ldots, 9, (,), +, \times\}, \{E, I\}, E, P)$ where P consists of the following production rules:

$$E \to E + E \mid E \times E \mid (E) \mid I, \quad I \to 0 \mid 1 \mid \ldots \mid 9 \mid 0I \mid 1I \mid \ldots \mid 9I$$

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Question: What does G generate?

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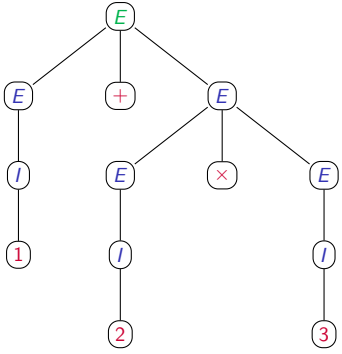
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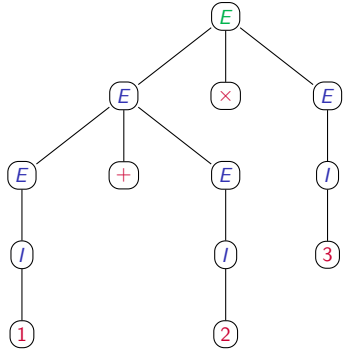
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Exercise

Find the two leftmost derivations of $1 + 2 \times 3$ corresponding to the two derivation trees of the previous slide.

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- Context-free grammars generate context-free languages, which are more general than regular languages.
- Derivation trees provide a graphical representation of derivations in context-free grammars.
- Leftmost and rightmost derivations are special types of derivations that are in one-to-one correspondence with derivation trees.
- Ambiguity in context-free grammars occurs when a string can be generated by multiple distinct derivation trees.

Next time

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Answer: Yes, they are called pushdown automata (PDA).